

Hypothesis

Research question: Do Persian-specific models reduce sub-word fragmentation and improve exact entity-boundary detection compared with XLM-R, especially for person names (PER)?

- **H1:** ParsBERT and TookaBERT produce fewer subwords per original word than XLM-R.
- **H2:** More fragmented PER mentions have more exact-boundary errors.

Background research:

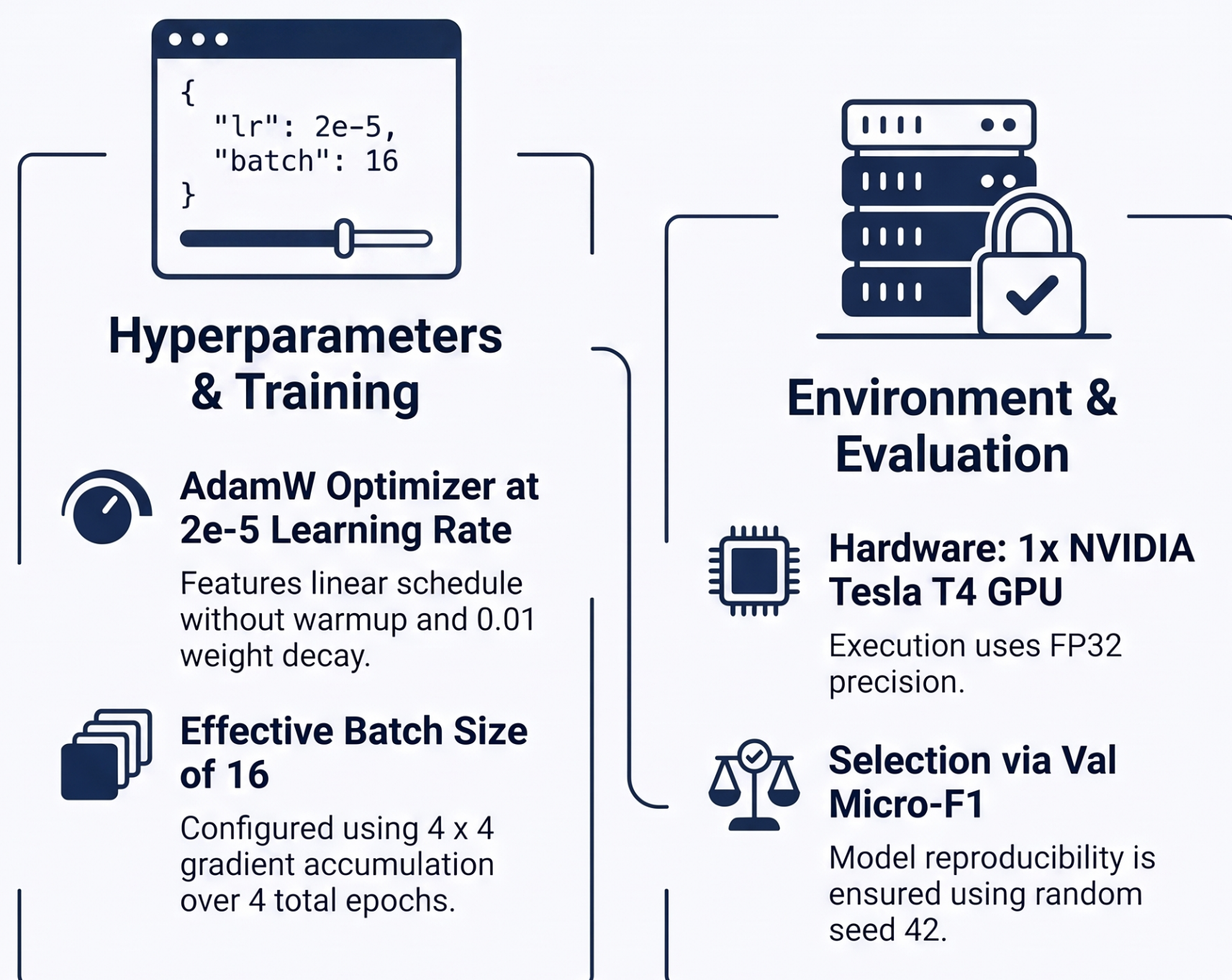
- In early works on the Persian language, Farahani and Ghassem-Sani evaluated mBERT + CRF on Arman and Peyma data. They introduced active learning strategies for low-resource training (Farahani & Ghassem-Sani, 2021). This is the baseline of this study.
- Ganjalipour et al. highlight XLM-R's SentencePiece subword tokenization as performing well for low-resource languages such as Persian. They customized XLM-R self-attention to help with Persian word order (SOV) and pro-drop properties by removing relative position clipping and adding PMI-based co-occurrence graph weights (Ganjalipour et al., 2023).
- SadraeiJavaheri et al. introduced TookaBERT-Base and TookaBERT-Large. TookaBERT's customized Byte Pair Encoding (BPE) vocabulary provides a useful baseline for comparing subword fragmentation rates (SadraeiJavaheri et al., 2024).

Our contribution: Explaining when fragmentation accompanies boundary errors and when it does not.

Methodology

- **Dataset:** Persian Named Entity Recognition (NER) contains: Person (PER), Location (LOC), Organization (ORG), FAC, EVENT, and PROD. Raw tags were mapped to a common scheme (PERS→PER, PRO→PROD) and checked for valid BIO transitions.
- **Models:** Three transformer models → each with different tokenization strategies:
 - ParsBERT (Persian-specific BERT)
 - TookaBERT-Base (Persian-specific BERT)
 - XLM-R-base (multilingual RoBERTa)

Model Training Setup at a Glance

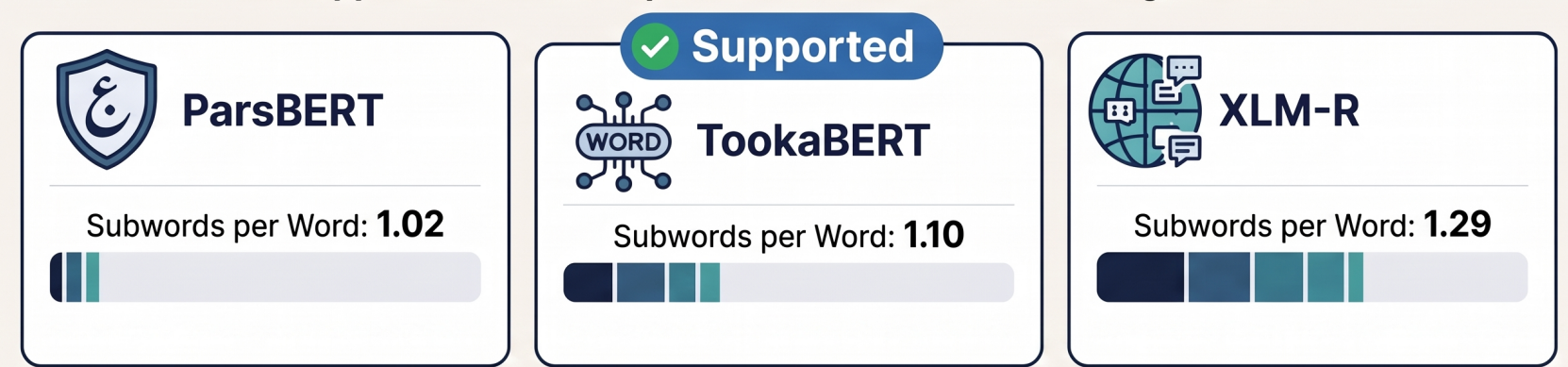


Discussion

Tokenizer Subword Fragmentation Analysis (H1)

Evaluation confirms Hypothesis 1 (H1): Persian-specific language models (ParsBERT and TookaBERT) demonstrate significantly lower subword segmentation per word than the general multilingual model (XLM-R).

H1 Supported: Persian-Specific Tokenizers Reduce Fragmentation



Results

Overall performance

Model	Typed micro F1	Macro F1	Bound. F1
XLM-R	.7988	.7148	.8514
ParsBERT	.8076	.7338	.8558
TookaBERT	.8111	.7285	.8606

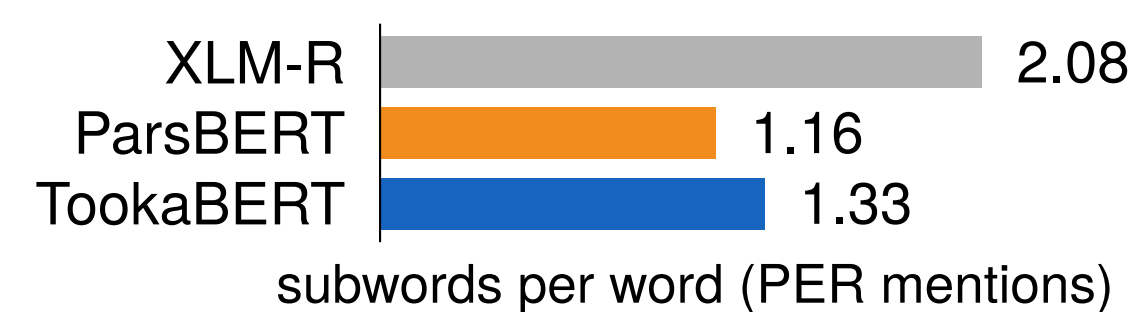
Person names (PER)

Model	Mean SFR	Bound. recall	Typed F1
XLM-R	2.0808	.9387	.9231
ParsBERT	1.1609	.9631	.9350
TookaBERT	1.3294	.9505	.9399

TookaBERT scores higher in overall typed and boundary F1; however, ParsBERT has the highest macro F1. The confidence intervals did not establish a clear difference in overall boundary detection between the models.

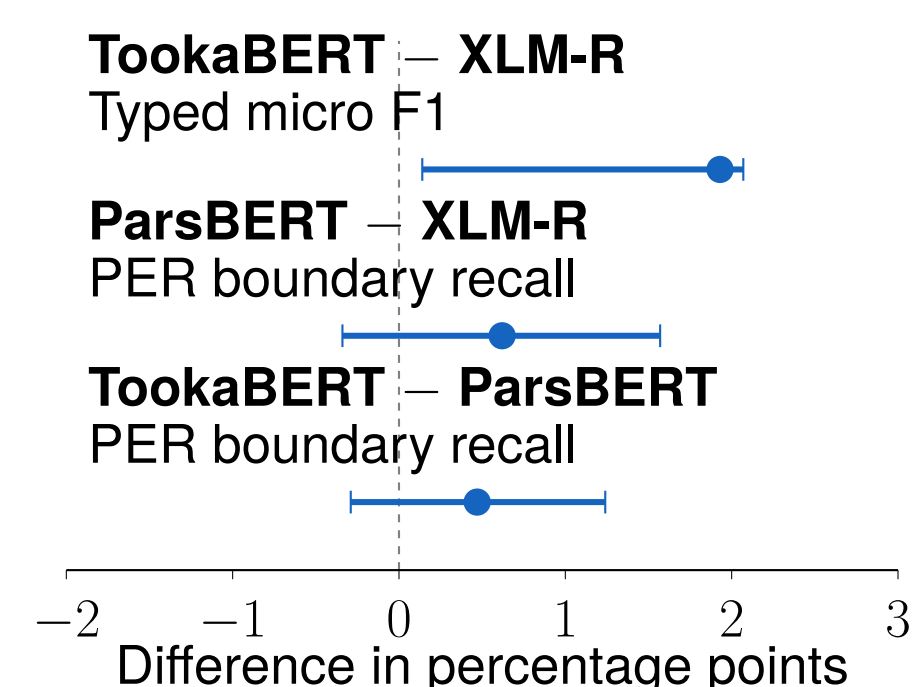
SFR is averaged over 1,110 gold "PER" mentions:

- ParsBERT recovers more exact PER boundaries.
- TookaBERT achieves higher typed PER F1.



Statistical uncertainty

95% paired bootstrap; differences in percentage points.



Dots show score differences; horizontal lines show 95% CIs. Positive values favour the first model.

Does fragmentation predict PER errors?

Model	Unfragmented	Fragmented
XLM-R	25/159 (15.72%)	42/951 (4.42%)
ParsBERT	3.81%	3.23%
TookaBERT	4.77%	5.34%

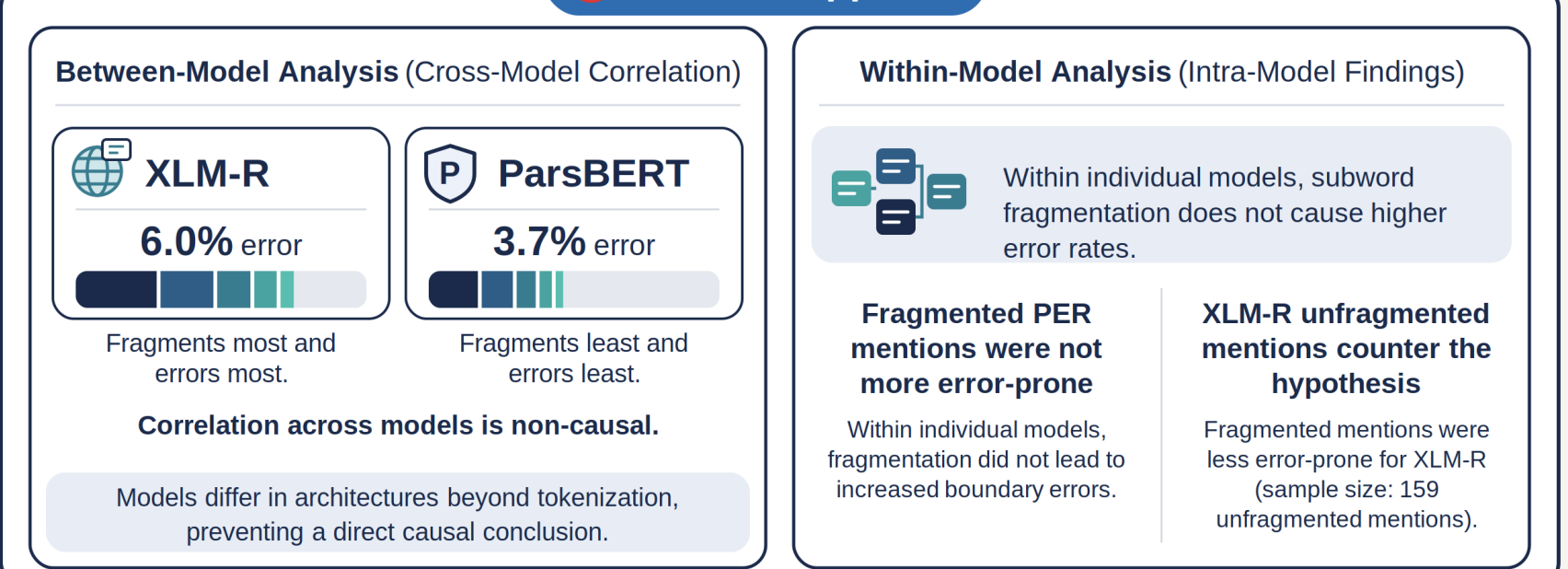
Greater fragmentation was not associated with higher boundary/error rates in these analyses.

Discussion

Tokenizer Fragmentation & PER Error Analysis (H2)

EVALUATION: Although cross-model comparisons show that higher-fragmentation models yield higher Person Entity Recognition (PER) boundary errors, within-model analysis proves that subword fragmentation itself does not cause higher error rates.

H2: Not supported



conclusion

- Fewer subwords and generally better NER: H1 supported.
- TookaBERT led in boundary F1; ParsBERT in PER boundary recall.
- Fragmentation did not consistently predict errors: H2 not supported.